

Soup DPO take-home | DON'T SHIP

Decision and run. DON'T SHIP. Evidence directory: evidence/t4-20261002T203632Z; run status: COMPLETED. Do not deploy this adapter. Synthetic heldout results cannot establish Russian customer-support quality. Missing evidence is UNRUN, never a pass.

Recipe and data. Qwen/Qwen2.5-1.5B-Instruct, immutable snapshot SHA recorded before training; Soup 0.75.2. T4/fp16, deterministic SDPA math; unquantized frozen base, streaming from RAM with 2 buffers, q_proj/v_proj LoRA r=8 alpha=16, AdamW, lr=5e-5, 1 epoch, pair batch 1, accumulation 4, max_length 512, seed 20260930. 500 template-defined fictional tickets authored with Codex: 400 train/100 heldout, grouped by scenario, 10 heldout groups. Shared policy/response templates remain a shortcut risk; no real tickets or independent labels.

Memory before vs after. First-order pre-run plan (not a hard bound): 3.815 GiB. Base 1,543,714,304 parameters (2.875 GiB resident alternative); host decoder store 2.441 GiB; LoRA 1,089,536 parameters. Two layer buffers 0.174, tied vocabulary 0.435, adapter+grad+Adam 0.016, activations 0.157, policy logits+backward 2.029, context/workspace/slack 1.000 GiB. Reference has no second base; sequential logits use max(policy, reference), not their sum. Frozen reference adapter allowance 0.004 GiB. Measured training allocated peak 2.017 GiB, reserved 6.439; gap vs plan -1.798 GiB. Plan assumes fully padded length 512 and conservative logit temporaries; real lengths/lifetimes differ. Reserved includes allocator slack; nvidia-smi adds driver/other processes and 1 s sampling misses short peaks. Audited train lengths 211-235 vs planned 512; post-hoc length-only plan 2.632 GiB. Main train command's sampled device peak 6.651 GiB (parity excluded). Device peak exceeded plan by 2.837 GiB. The fixed 1 GiB cache/context allowance is not a hard bound; reserved blocks can exceed it. Peaks are not simultaneous; no single-cause attribution is proved.

Prove training. Falling loss may reflect sample order/length, reference drift, label bugs or training tensors that never load into the serving model. Part 2 compares saved tensors to their exact initialized snapshot (not to zero); requires 90% of LoRA B matrices to change, all finite. Ordinary PEFT reload must round-trip every adapter tensor with no missing-key warnings; identical response-token log-probabilities must change beyond repeated-base noise and return to base when the adapter is disabled. Observed passed=True; changed B fraction=1.000; functional delta=11.1149, noise=0. Actual optimizer steps=100; reference delta=0. T4 streamed-vs-resident DPO backward: True; lr=0 control: correctly rejected. Tensor/output checks miss wrong gradients and bad labels; parity covers three batches at this stack/model/shape, not all steps.

Evidence, failure modes and limits

Silent failures: check and Soup coverage. Inert saved adapter: Round-trip keys, tensor deltas, enabled/disabled probes; data doctor/preflight cannot prove it. Historical .inner. defect is fixed upstream; our saved artifact has an independent reload check. Wrong streamed gradients: Compare all LoRA gradients to resident DPO with deliberately nonzero B. Forward/loss equality is insufficient. NF4 aliasing defect does not apply to our quantization=none; no NF4 correctness claim. Reference drift / extra model: Capture ref logs before/after; record ref instance/ref-adapter and runtime buffers. Same frozen base must be reused. Soup's fit gate does not establish a stationary reference. Masking, EOS, truncation: Run doctor on chat projection, lint on actual DPO pairs; audit all 500 tokenized pairs and inspect prepared TRL IDs. Chat doctor does not prove the DPO trainer's masking. Duplicates / length shortcuts: Group split, exact-prompt checks, sum and mean margins; lint catches heuristics. Shared synthetic templates and label validity remain unresolved, even if lint is OK. Precision / skipped steps / OOM: Require T4 fp16, materialized LoRA, finite nonzero gradients, step count and saved-state change. Raw CUDA peaks plus nvidia-smi. Preflight sizes fit, not learning; measured SFT probe is not DPO evidence. Misleading SHIP: Run Soup offline on measured preference accuracy and explicitly named four-question forced-choice sanity set. A numeric SHIP can miss label correctness, confidence, real-domain generalization and adapter provenance.

Verdict and required changes. Soup ship output: recorded in soup_ship.json. Numeric gate returned SHIP; this does not override the final DON'T SHIP call. Synthetic paired accuracy 0.87 to 1.00; Heldout paired accuracy delta +0.130; group-bootstrap 95% interval [0.0, 0.36]. Soup near-duplicate lint was skipped (datasketch absent); chat doctor reported missing generation markers (MINOR). Exact/group checks and actual TRL collator tests are separate evidence. Required before SHIP: retain a passing T4 evidence chain; obtain independently annotated real Russian tickets with scenario separation; manually inspect generated replies and safety errors; run broader Russian/general regression tests and seeds; resolve every MAJOR diagnostic. Package this run's weights with its snapshot/config/data hashes. No failed run is removed.

Surprise and remaining concern. Default-attention T4 probes had identical losses but disagreeing gradients; even resident repeats varied. Disabling fp16 GEMM reductions alone did not resolve this. Deterministic SDPA math made all 112 gradients bit-identical on three batches, without relaxing tolerances. This numerical variability confounded the original streaming test; it did not establish a streaming defect. My remaining concern is that shared synthetic templates reward phrasing instead of real support quality; neither loss nor four general probes can settle that.

AI assistance. Codex prepared the dataset templates, implementation, documentation and source analysis. Accepted suggestions: immutable base snapshot, actual initialization comparison, ordinary PEFT reload, lr=0 negative control and backward parity. Corrected by checking source: doctor rejects DPO pairs; use lint plus chat projection; T4 precision is fp16; never report pending GPU checks as passed. Local tests and T4 evidence must be distinguished. The applicant has not independently reviewed or run this work unless their review is separately recorded; no applicant verification is invented.